Differential Equation: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Basic terms

KEY POINTS

- DE: relation of x, y and derivatives of y; ordinary = one independent variable, partial = two or more
- General solution (complete integral / primitive): number of independent constants = order
- Particular solution: constants fixed by the given condition $y(x_0) = y_0$
- A condition like "y bounded" or "$y \to 0$ as $x \to \infty$" also fixes a constant (usually makes the Ce^{kx} term vanish)

2. Order and degree

KEY POINTS

- Order = highest derivative present
- Degree = power of the highest derivative after making the DE a polynomial in derivatives (clear radicals, fractional powers)
- Degree not defined if a derivative stays inside sin, $e^{(\ )}$, ln; but $\sin^{-1} y' = x \Rightarrow y' = \sin x$ has degree 1

Watch out

Order and degree are positive integers; remove radicals of derivatives before reading the degree

3. Formation of a DE

STEPS

1. Write the family with the fewest independent constants ($c_1 e^{x+c_2} = Ae^x$)
2. Differentiate as many times as there are constants
3. Eliminate the constants; order of DE = number of constants

KEY POINTS

- Family in words: write its equation first (circles touching x-axis at O: $x^2 + y^2 - 2ay = 0$)
- $y^2 = 4A(x + A)$ has 1 constant; $y^2 = 4A(x + B)$ has 2

4. Variable separable

Formulas

$$f(x)\,dx = g(y)\,dy \;\Rightarrow\; \int f(x)\,dx = \int g(y)\,dy + c$$

KEY POINTS

- Take $+\ln c$ when the other terms are logarithms
- $\frac{dy}{dx} = f(ax + by + c)$: put $t = ax + by + c$, $\frac{dt}{dx} = a + bf(t)$
- $\frac{dy}{dx} = \frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}$, $a_2 + b_1 = 0$: cross-multiply, group $d(xy)$

Watch out

Do not lose solutions when dividing (e.g. $y = 0$, $y = 3$ constant solutions)

5. Polar substitutions

Formulas

$$x = r\cos\theta,\ y = r\sin\theta:\ x\,dx + y\,dy = r\,dr,\quad x\,dy - y\,dx = r^2 d\theta$$

$$x = r\sec\theta,\ y = r\tan\theta:\ x\,dx - y\,dy = r\,dr,\quad x\,dy - y\,dx = r^2\sec\theta\,d\theta$$

KEY POINTS

- Use when $x\,dx \pm y\,dy$, $x\,dy - y\,dx$ appear with $x^2 \pm y^2$ or $\frac{y}{x}$

6. Curves from geometric conditions

Formulas

Tangent: $Y - y = m(X - x)$, x-int $= x - \frac{y}{m}$, y-int (initial ordinate) $= y - mx$
Normal: $Y - y = -\frac{1}{m}(X - x)$, x-int $= x + my$
Subtangent $\left|\frac{y}{m}\right|$, subnormal $|ym|$, lengths $\left|\frac{y}{m}\right|\sqrt{1+m^2}$, $|y|\sqrt{1+m^2}$

STEPS

1. Take $P(x, y)$ on the curve, $m = \frac{dy}{dx}$
2. Translate the condition into a DE (polar radius $OP = \sqrt{x^2 + y^2}$)
3. Solve; use the given point; check both signs of $\pm$

KEY POINTS

- Area or arc-length condition $\int_a^x \ldots$: differentiate both sides first

7. Integral identities

STEPS

1. Differentiate the identity (Leibniz rule) to get a DE
2. Put $x =$ lower limit in the identity for the initial value
3. Solve the DE (separable or linear)

8. Homogeneous DE

Formulas

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right) : \; y = vx, \quad \int \frac{dv}{F(v) - v} = \ln|x| + c$$

KEY POINTS

- Test: $f(\lambda x, \lambda y) = \lambda^n f(x, y)$ for numerator and denominator with the same n
- If $\frac{dx}{dy} = G\left(\frac{x}{y}\right)$ is simpler, put $x = vy$

9. Reducible to homogeneous

STEPS

1. $\frac{dy}{dx} = \frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}$: if $a_2 + b_1 = 0$, group into $d(xy)$
2. If $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ (parallel), put $t = a_1x + b_1y$
3. Else shift to the intersection (h, k): $x = X + h$, $y = Y + k$, then $Y = vX$

10. Linear DE

Formulas

$$\frac{dy}{dx} + Py = Q : \quad \text{IF} = e^{\int P\,dx}, \quad y \cdot \text{IF} = \int Q \cdot \text{IF}\,dx + C$$

$$\frac{dx}{dy} + P_1x = Q_1 : \quad x \cdot e^{\int P_1 dy} = \int Q_1 e^{\int P_1 dy}\,dy + C$$

KEY POINTS

- Linear: y and derivatives only in degree one, never multiplied; general form $a_0y^{(n)} + \cdots + a_ny = \phi(x)$
- Linear $\Rightarrow$ degree 1, but degree 1 $\not\Rightarrow$ linear
- y non-linear but x linear: use the $\frac{dx}{dy}$ form
- Inequality $f' + Pf > 0$ (e.g. $f' > 2f$): $(f \cdot \text{IF})' > 0$, so $f \cdot \text{IF}$ is increasing

Watch out

Make the coefficient of $\frac{dy}{dx}$ equal to 1 before finding IF; $e^{\ln g} = g$, $e^{-\ln g} = \frac{1}{g}$

11. Bernoulli and change of variable

Formulas

$$y' + Py = Qy^n : \ z = y^{1-n}, \quad z' + (1-n)Pz = (1-n)Q$$
$$f'(y)\,y' + Pf(y) = Q : \ t = f(y), \quad t' + Pt = Q$$

KEY POINTS

- Typical $f(y)$: $\tan y$ (with $\sec^2 y$), e^y, $\ln y$ (with $\frac{1}{y}$), $\sin y$ (with $\cos y$), y^2 (with $2y$)

12. Rate problems (growth, decay, cooling, mixture)

Formulas

$\frac{dN}{dt} = kN \Rightarrow N = N_0 e^{kt}$; cooling $\frac{dT}{dt} = -k(T - T_s) \Rightarrow T - T_s = (T_0 - T_s)e^{-kt}$
Mixture: $\frac{dQ}{dt} =$ (rate in) $- \frac{Q}{V(t)} \cdot$ (outflow rate)

STEPS

1. Name the variable and write "rate = ..." as a DE
2. Use two given data to find the constant and k

13. Orthogonal trajectories

STEPS

1. Form the DE of the family (eliminate the parameter)
2. Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$
3. Solve the new DE

KEY POINTS

- Isogonal: fixed angle α; orthogonal: $\alpha = \frac{\pi}{2}$

14. Exact differentials

Formulas

$$x\,dy + y\,dx = d(xy), \quad \frac{x\,dy - y\,dx}{x^2} = d\left(\frac{y}{x}\right), \quad \frac{y\,dx - x\,dy}{y^2} = d\left(\frac{x}{y}\right)$$
$$\frac{x\,dy + y\,dx}{xy} = d(\ln xy), \quad \frac{x\,dy - y\,dx}{xy} = d\left(\ln\frac{y}{x}\right), \quad \frac{x\,dy - y\,dx}{x^2+y^2} = d\left(\tan^{-1}\frac{y}{x}\right)$$
$$\frac{x\,dx + y\,dy}{x^2+y^2} = d\left(\frac{1}{2}\ln(x^2+y^2)\right), \quad \frac{x\,dy + y\,dx}{x^2y^2} = -d\left(\frac{1}{xy}\right), \quad \frac{ye^x\,dx - e^x\,dy}{y^2} = d\left(\frac{e^x}{y}\right)$$

KEY POINTS

- Divide by a suitable factor (x^2, y^2, xy, x^2+y^2) and group; $yf(xy)\,dx + xg(xy)\,dy = 0$: put $xy = v$

15. Common mistakes checklist

STEPS

1. Degree asked: make the DE a polynomial in derivatives first
2. Linear DE: coefficient of y' must be 1 before IF
3. Check the domain sign: $e^{\int \frac{dx}{x}} = |x|$, take x for $x > 0$
4. Geometric problems: keep both signs of $\pm$ until the point decides
5. Do not forget the constant; apply the initial condition only after integrating